

Direct detachment from ring-like cores

find_ring_substituent_cores — MolDAM's addition, no MolPLA analogue

RECAP's fragment tree cannot cut a bond it has no rule for, so a molecule whose only detachable substituents hang off a ring yields no core at all. This path cuts every non-ring SINGLE bond, keeps the split where the larger side clears ratio x nHA, and takes that side as the core.

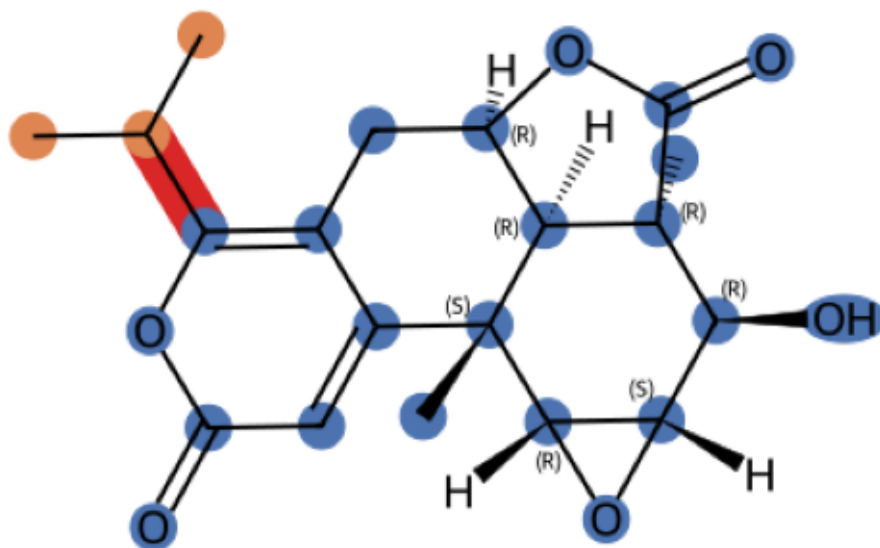
In MolPallate it is a FALLBACK: used only when the fragment tree returns nothing, because its cores are single-R-group by construction and mixing them in wholesale pushed $k \geq 2$ from 60.8% down to 36.7%.

Measured on 15,000 COCONUT molecules at ratio 1/3, min_rgroup_atoms 2:

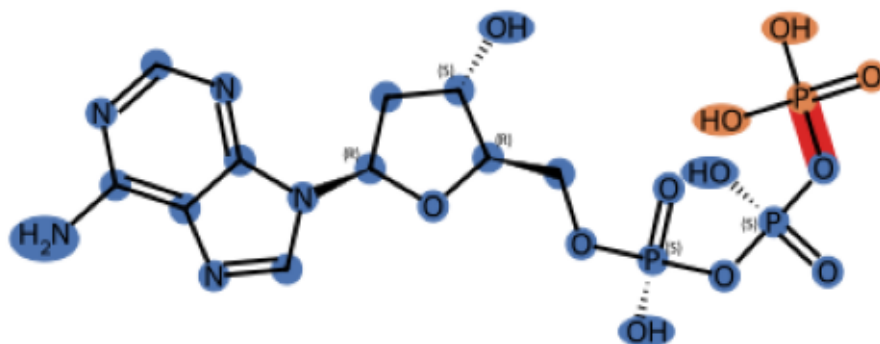
fragment-tree path	79.1%	
ring-substituent path	17.2%	<- rescued by this fallback
no decomposition	3.7%	

Blue = core, orange = R-group, red = the cut bond.

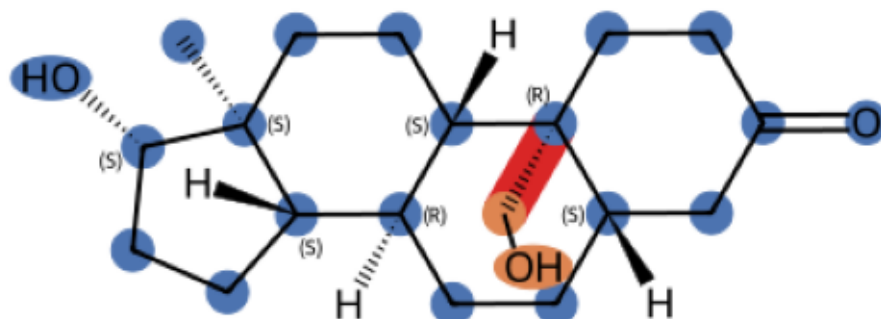
CC(C)c1oc(=O)cc2c1C[C@H]1OC(=O)[C@@]3(C)[C@@H](O)[C@@H]4O[C@@H]4O[C@@H]3C
 core 22 atoms (17 in rings) | 1 R-group(s): CCC



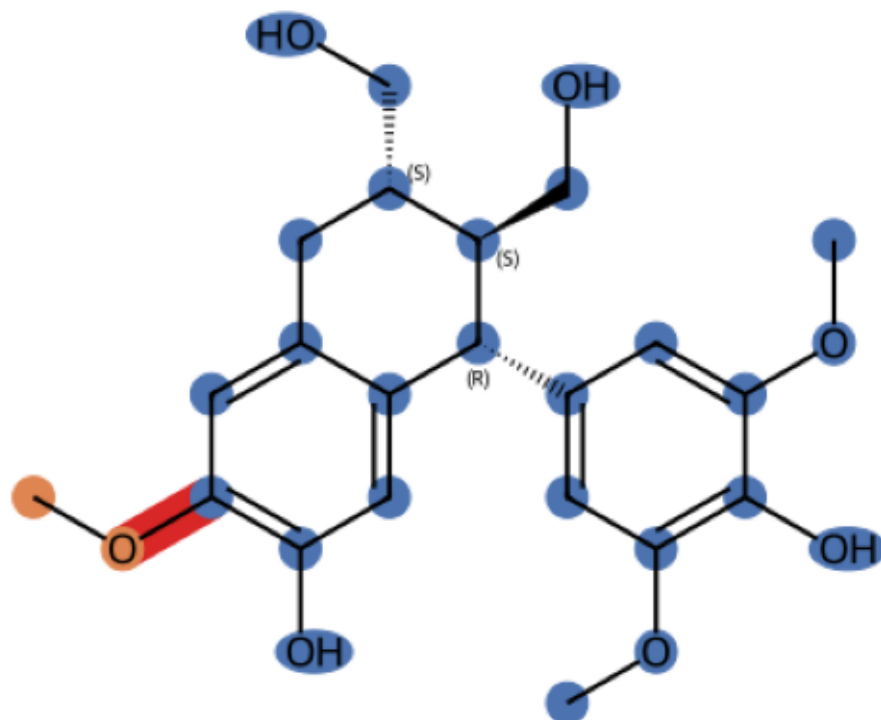
Nc1ncnc2c1ncn2[C@H]1C[C@H](O)[C@@H](CO[P@](=O)(O)O[P@](=O)(O)O)O1
 core 26 atoms (14 in rings) | 1 R-group(s): O=P(O)O



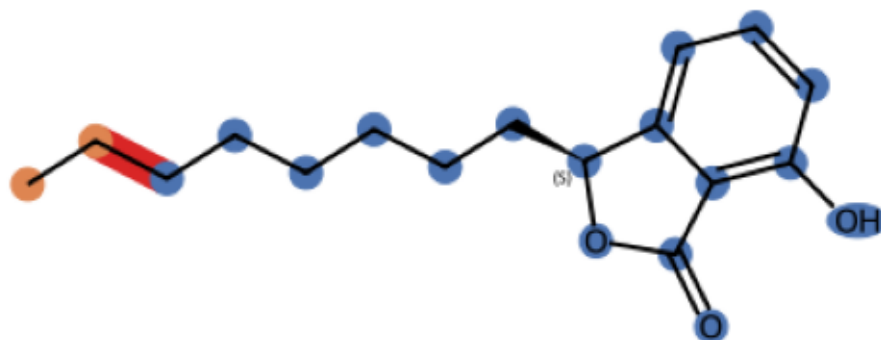
C[C@]12CC[C@H]3[C@@H](CC[C@H]4CC(=O)CC[C@@]43CO)[C@@H]1CC[C@@H]
 core 20 atoms (17 in rings) | 1 R-group(s): CO



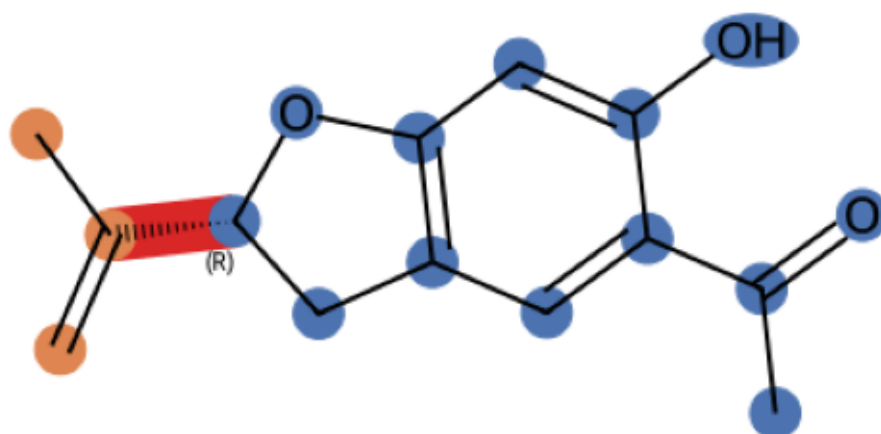
COC1cc2c(cc1O)[C@@H](c1cc(OC)c(O)c(OC)c1)[C@H](CO)[C@@H](CO)C2
 core 26 atoms (16 in rings) | 1 R-group(s): CO



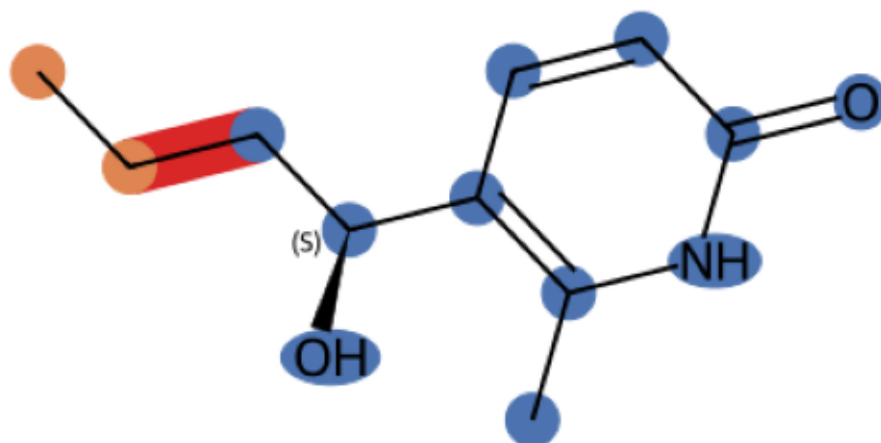
CCCCCCC[C@@H]1OC(=O)c2c(O)ccc21
 core 17 atoms (9 in rings) | 1 R-group(s): CC



C=C(C)[C@H]1Cc2cc(C(C)=O)c(O)cc2O1
 core 13 atoms (9 in rings) | 1 R-group(s): C=CC



CCC[C@H](O)c1ccc(=O)[nH]c1C
 core 11 atoms (6 in rings) | 1 R-group(s): CC



O=c1[nH]c(=O)c2nc(Br)n([C@H]3O[C@@H](CO)[C@H](O)[C@@H]3O)c2[nH]
 core 19 atoms (14 in rings) | 1 R-group(s): CO

